Internet of Things and Vehicular Networks (TP-546)

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Agriculture 4.0: Traceability

Agriculture 4.0: Traceability

- What is traceability in Agriculture 4.0?

- Definition: the ability to track the movement of agricultural products throughout the supply chain, from farm to table.

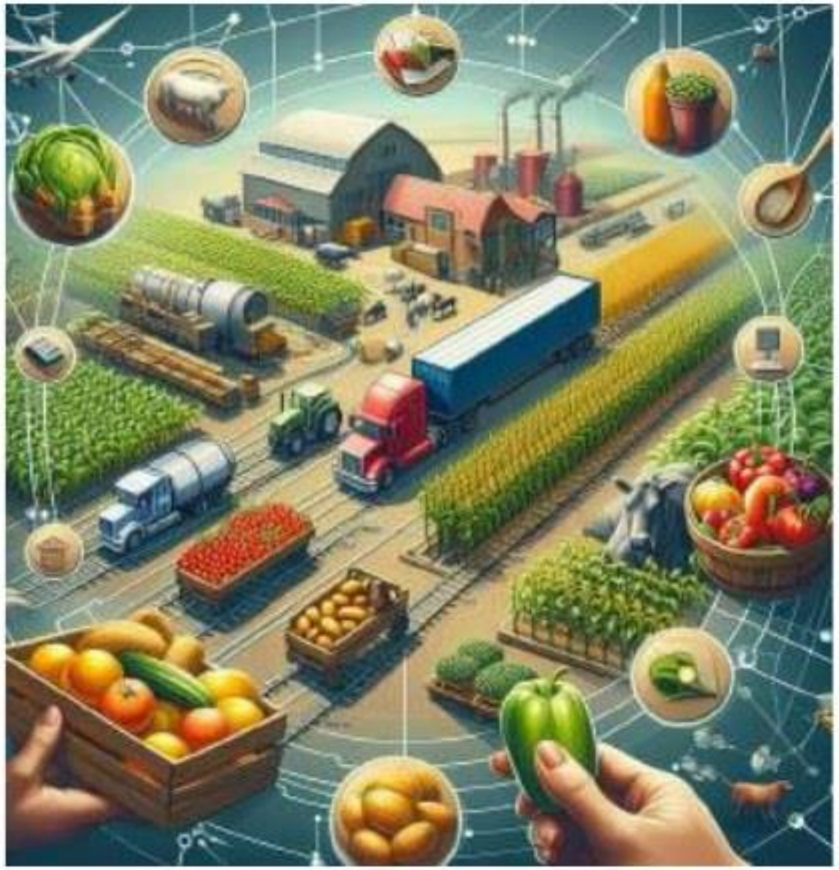
Agriculture 4.0: Traceability

• Traceability is the ability to track the origin, movement, and transformation of products across every stage of production, processing, and distribution.

Agriculture 4.0: Traceability

• Traceability builds consumer trust by providing transparent information about a product's journey.

• It supports food safety, quality control, and compliance with regulatory requirements.



Agriculture 4.0: Traceability

- ISO 22005:2007 defines principles and requirements for food and feed traceability systems. It helps organizations:

1. Track materials, ingredients, food, feed, and packaging;
2. Identify the documentation required at each production stage;
3. Coordinate actors across the chain;
4. Improve communication among stakeholders;
5. Improve information reliability, effectiveness, and productivity.

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• Food safety is a global concern because of recurring safety incidents.

• African swine fever, avian influenza, bovine spongiform encephalopathy, foot-and-mouth disease, and microorganisms such as Salmonella have increased public and private attention to food attributes.

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- Effective traceability systems significantly reduce response time during animal- or plant-disease outbreaks by providing rapid access to reliable information about the source and location of affected products.

Agriculture 4.0: Traceability

- Traceability strengthens prevention instead of relying only on reaction to food-safety violations.
- Supported by ICT, traceability systems help companies monitor and manage risks in real time.
- Better information also supports management decisions, market access, and lower operating costs.

Agriculture 4.0: Traceability

- Traceability visibility helps companies use resources and processes more effectively, improving long-term profitability.

- Proper implementation can reduce obsolete-product losses and inventory levels, accelerate the identification of supplier or process problems, and improve logistics and distribution.

Agriculture 4.0: Traceability

• Greater customer confidence strengthens brands and brand value.

• For premium products such as saffron, vanilla, cloves, cocoa, and other spices, traceability prevents contamination or mixing with lower-value goods and helps guarantee authenticity.

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- Basic characteristics of traceability systems:

- Identification of units and batches for every ingredient and product;
- Records of when and where units or batches move or are transformed;
- A system that links these records and transfers relevant traceability data with the product to the next processing stage.

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- Main factors affecting traceability effectiveness:

- Supply-chain structure and organization;
- Collaboration among supply-chain actors;
- Number of actors with internal and external traceability;
- Ability to identify product origin;
- Ability to manage traceability systems;
- Compatibility among participants.

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- Main factors affecting traceability effectiveness:

- Product destination;
- Identification of the traceable batch unit;
- Time required to trace a product;
- Credibility of the traceability method.

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- Main factors affecting traceability effectiveness:

- Data-identification methods and data standards;
- Integration with existing information-management and quality- or safety-assurance systems;
- Traceability legislation.

Agriculture 4.0: Traceability

- Key components:
- Data collection and management using sensors and IoT devices.
- Source: https://maritima.com/en/perishable-goods/

Agriculture 4.0: Traceability

Temperature sensors monitor conditions inside the food truck and ensure perishable items remain at safe temperatures.

Proximity sensors detect nearby objects or people and can prevent door collisions or activate lighting.

Gas sensors detect hazardous gases such as propane or carbon monoxide; ventilation and leak detection are essential for safety.

Agriculture 4.0: Traceability

- Motion sensors detect movement for security, such as unauthorized entry, or to activate lights when someone enters the truck.

- Smoke detectors identify smoke particles and trigger alarms, helping prevent fires and protect occupants and the vehicle.

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• Light sensors adjust illumination according to ambient conditions, saving energy by activating lights only when needed.

• Door and window sensors report whether openings are closed. They support security and climate control, such as switching off air conditioning when a door is open.

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- Key components:
- Unique identification using barcodes, QR codes, and RFID tags.

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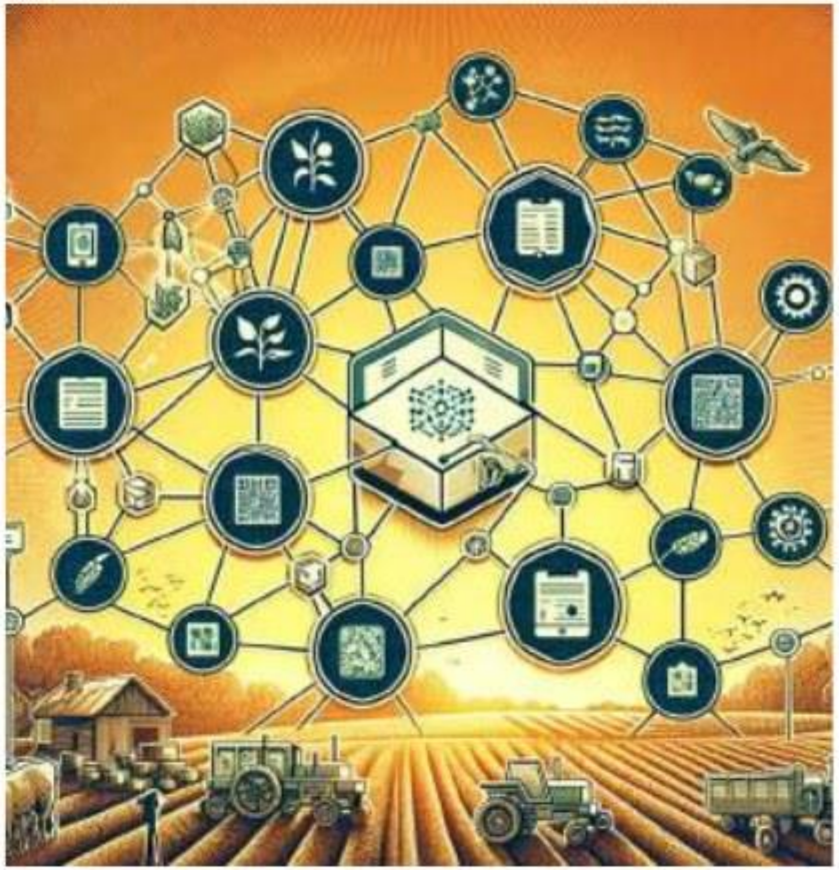
- Key components:

- Digital platforms for data sharing and analysis;
- Blockchain technology for stronger security and transparency.



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- Challenges and considerations:

- Implementation cost may be a barrier for small producers;
- Data privacy and security are critical;
- Common standards are needed for data collection and sharing;
- Limited rural connectivity can hinder digital traceability solutions.

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- Traceability is a major enabler of Agriculture 4.0.
- Producers can increase efficiency, ensure food safety, and gain a competitive advantage.
- Consumers gain transparency and can make informed food choices.
- Traceability supports a more sustainable and transparent food system.

Agriculture 4.0: Traceability

In agriculture, blockchain creates an immutable record of every stage in the supply chain.

Agriculture 4.0: Traceability

• How blockchain improves traceability in Agriculture 4.0:

• Immutable records: planting, harvesting, processing, transportation, and sales are recorded as tamper-resistant blocks that cannot be erased or changed.

• Transparency: authorized participants access the same information, increasing visibility and trust.

Agriculture 4.0: Traceability

• How blockchain improves traceability in Agriculture 4.0:

• Improved food safety: contamination or quality issues can be traced rapidly, enabling targeted recalls and lower risk.

• Efficiency and cost reduction: blockchain reduces intermediaries and manual data entry, streamlining processes across the supply chain.

Agriculture 4.0: Traceability

• Proof of origin and authenticity: consumers can verify where products came from, supporting fair trade and ethical producers.

• Sustainability and compliance: blockchain can track sustainable practices, certifications, and regulatory compliance, promoting responsible agriculture.

Agriculture 4.0: Traceability

- Real-world examples of Agriculture 4.0 traceability:

- IBM Food Trust;
- AgTrace;
- Wholechain;
- Brazil's ConectarAGRO initiative.

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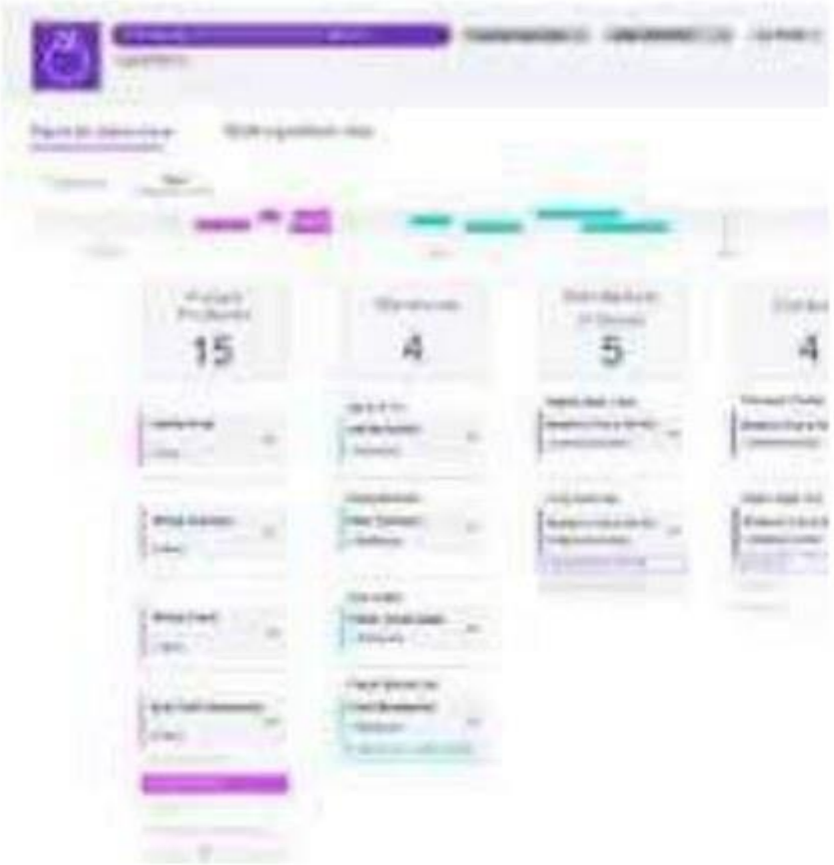
• IBM Food Trust is a blockchain-based platform designed to improve visibility and accountability throughout the food supply chain.

• It connects producers, processors, distributors, manufacturers, retailers, and other participants in a collaborative network.

Agriculture 4.0: Traceability

- How it works:

- Blockchain technology: IBM Food Trust creates a secure, shared, permissioned record of food-system data.



Agriculture 4.0: Traceability

- Data sharing and access: network participants share and access relevant data according to their roles and permissions.

- This improves transparency and traceability from farm to plate.

Agriculture 4.0: Traceability

- How it works:

- Improved visibility: IBM Food Trust allows users to trace food journeys rapidly, identifying origins, processing sites, and distribution points. Potential contamination or fraud can therefore be investigated more efficiently.

Agriculture 4.0: Traceability

- Benefits:
- Improved food safety: stronger traceability helps identify and address safety risks rapidly.
- Reduced food waste: identifying bottlenecks and inefficiencies minimizes waste and improves resource use.

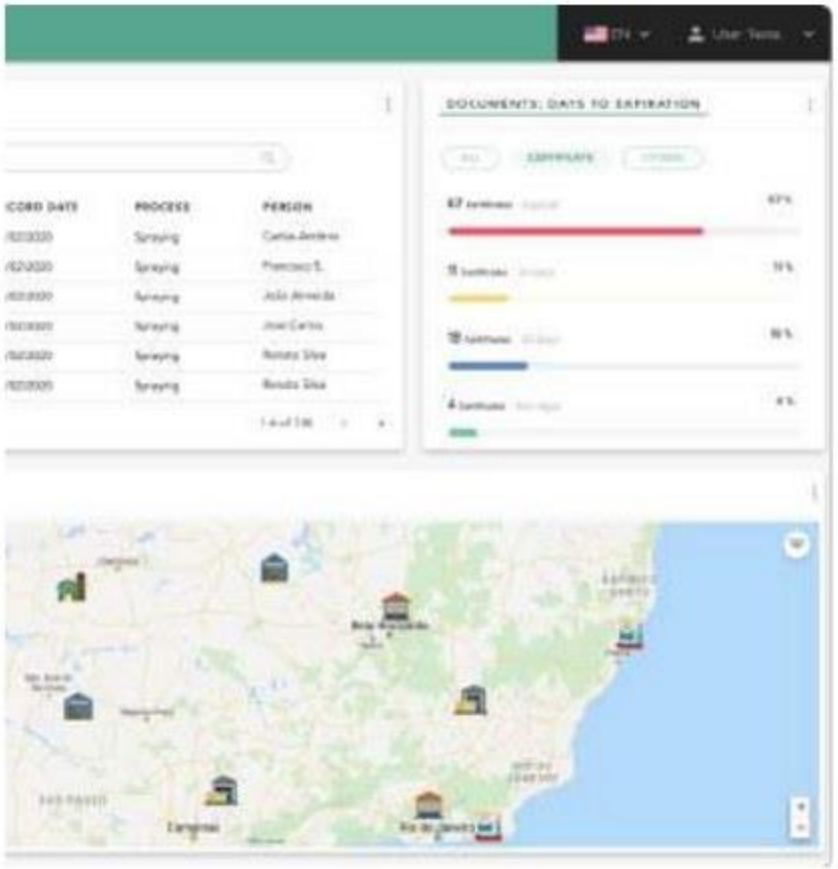
Agriculture 4.0: Traceability

- Benefits:
- Greater consumer confidence: transparency about origin and product journeys strengthens trust in the food system.
- Supply-chain optimization: simplified data sharing and better visibility improve efficiency and reduce costs.

Agriculture 4.0: Traceability

- AgTrace is a traceability solution designed specifically for agriculture, with a focus on livestock.

- It provides a comprehensive system for tracking animals and animal products throughout the supply chain.



Agriculture 4.0: Traceability

- AgTrace features and benefits:
- End-to-end traceability tracks animals from birth through breeding, transport, processing, and distribution.
- RFID tags, barcodes, and sensors collect and manage identification, health, feeding, and other relevant data.

Agriculture 4.0: Traceability

• AgTrace features and benefits:

• Real-time monitoring gives stakeholders immediate access to the movement and condition of animals and products across the supply chain, supporting early problem detection and timely action.

Agriculture 4.0: Traceability

• Wholechain is a blockchain-based traceability platform designed to bring trust, coordination, and transparency to fragmented supply chains.

• It tracks products from origin to consumer, supporting visibility and accountability at every stage.

Wholechain®

Agriculture 4.0: Traceability

- Use cases:

- Food and agriculture: Wholechain tracks origin, processing, and distribution to support safety and sustainable practices.

- Seafood: the platform has been used to fight illegal fishing and promote responsible sourcing.

Agriculture 4.0: Traceability

• Use cases:

• Cosmetics and apparel: Wholechain verifies sustainability and fair-trade claims, supporting ethical sourcing.

• Manufacturing: the platform tracks raw materials and components, supporting quality control and regulatory compliance.

Agriculture 4.0: Traceability

- Brazil's ConectarAGRO initiative is a collaborative effort to expand 4G internet coverage in rural areas.

- Recognizing that connectivity drives agricultural development, it gives farmers digital tools and resources to improve productivity, sustainability, and economic growth.

Agriculture 4.0: Traceability

• Main objectives:

• Provide affordable and practical internet access to farmers and agricultural communities.

• Promote precision agriculture by enabling sensors, drones, and data analytics for more efficient resource management and optimized yields.

Agriculture 4.0: Traceability

- Main objectives:

- Empower farmers with information, market data, financial services, and educational resources;
- Encourage innovation through technologies and services tailored to rural communities;
- Support sustainable practices, environmental protection, resource conservation, and climate resilience.

Agriculture 4.0: Traceability

- Main objectives:

- Empower farmers by providing access to information, market data, financial services, and educational resources, enabling informed decisions and improved livelihoods.

Agriculture 4.0: Traceability

- Main objectives:

- Foster innovation: by creating a connected ecosystem, ConectarAGRO encourages technologies and services tailored to rural communities.

Agriculture 4.0: Traceability

• Expected impact:

• Reliable internet access can transform Brazilian agriculture by enabling precision technologies and data-driven decisions that increase productivity and efficiency.

Agriculture 4.0: Traceability

- ConectarAGRO is a significant step toward a connected and digitally enabled Brazilian agricultural sector.
- By narrowing the digital divide, it can unlock greater productivity, sustainability, and economic growth nationwide.

Use Cases

• IBM Food Trust and Walmart:

• Walmart partnered with IBM to use the IBM Food Trust blockchain platform to track food from farm to shelf.

• The traceability system is based on Hyperledger Fabric.

Use Cases

- IBM Food Trust and Walmart:
- One pilot traced mangoes sold in U.S. stores; another traced pork sold in China.
- For Chinese pork, authenticity certificates were uploaded to the blockchain.
- For U.S. mangoes, origin-tracing time fell from seven days to 2.2 seconds.

Use Cases

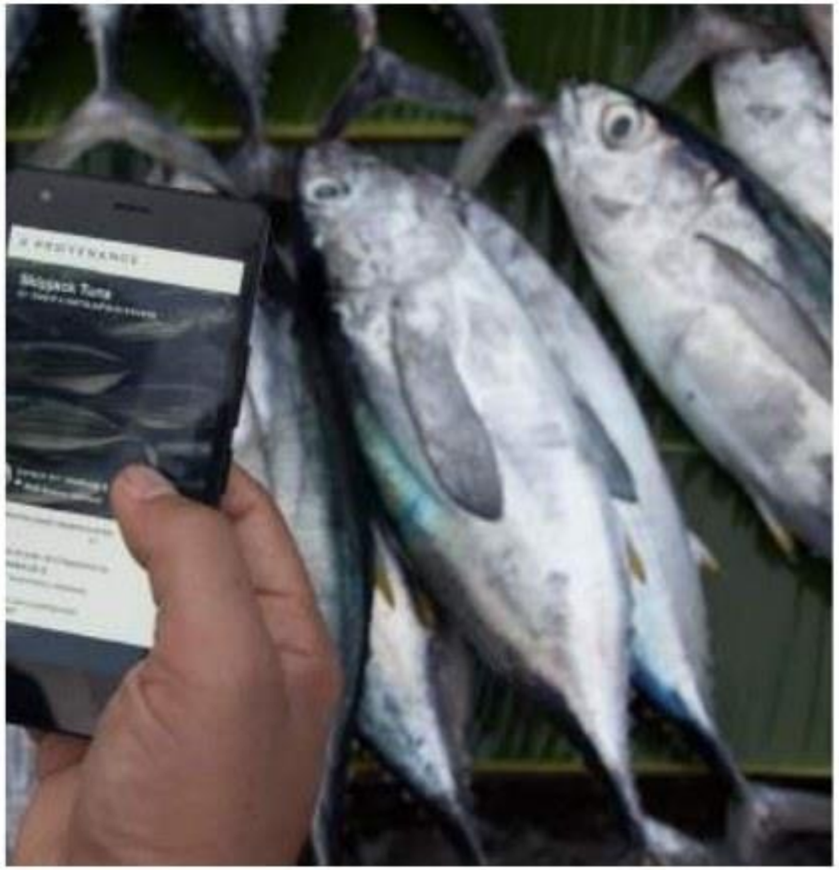
- IBM Food Trust and Walmart:
- Walmart can track the origin of more than 25 products from five suppliers using Hyperledger Fabric.
- Fresh leafy-green suppliers are required to trace products such as lettuce and spinach through the system.

Use Cases

- Provenance:

- Provenance uses blockchain to track the origin and journey of food products.

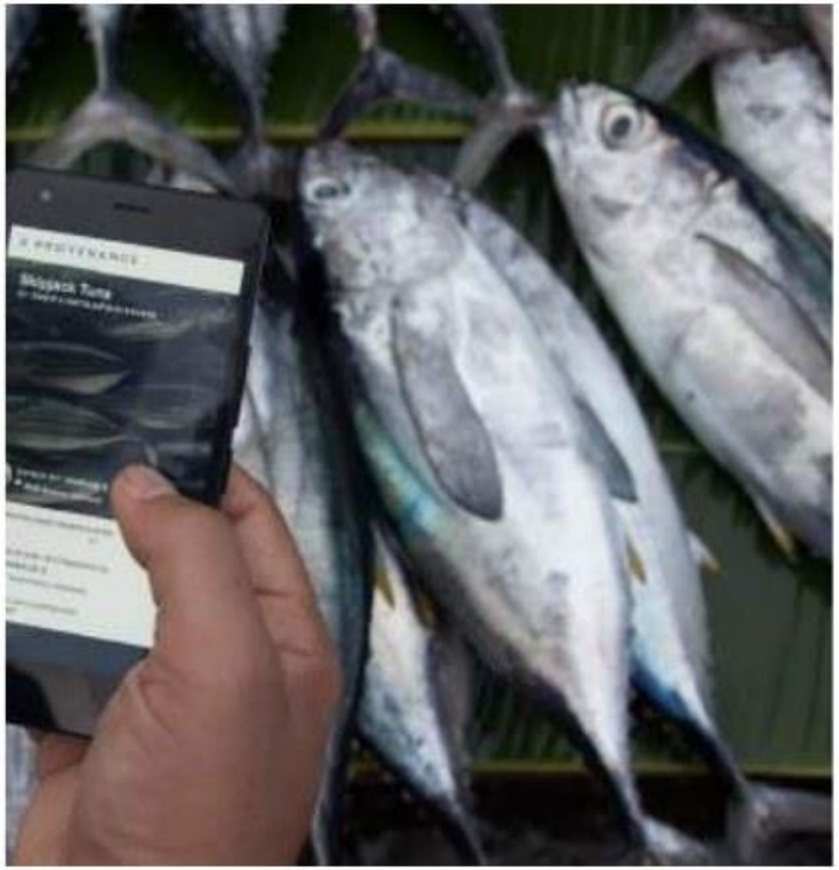
- Indonesia is the world's largest tuna-producing country, and tuna fishing is an important source of employment and foreign income.



Use Cases

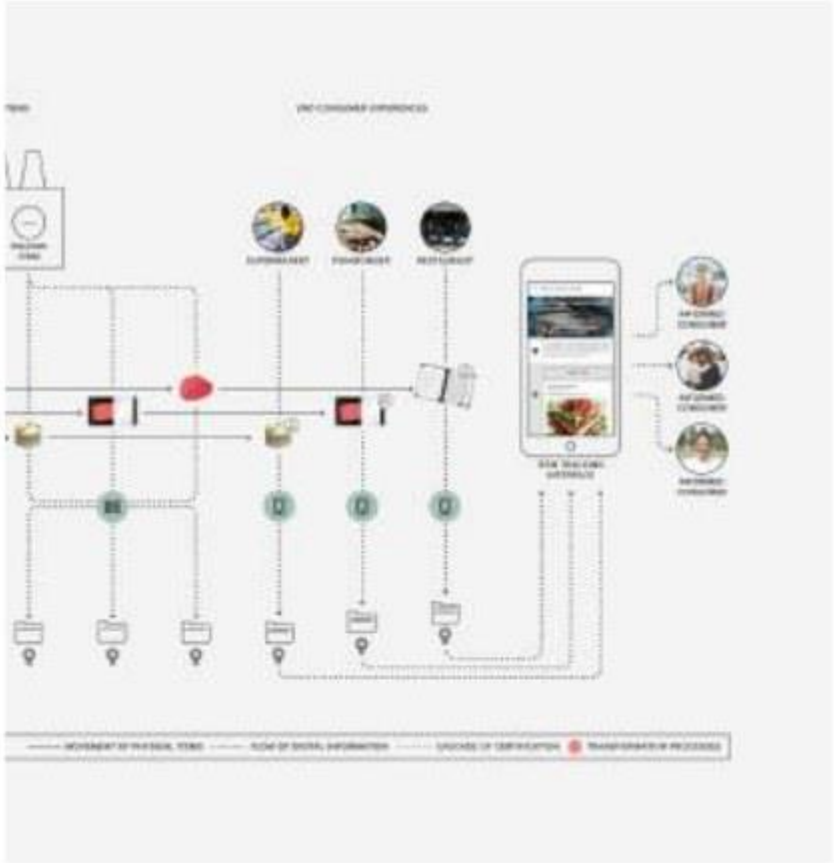
• The tuna industry faces overfishing, human-rights abuses, fraud, and illegal, unreported, and unregulated fishing.

• Provenance tracks tuna from Indonesian waters to the final consumer, helping confirm sustainable harvesting and processing.



Use Cases

- Source: https://www.provenance.org/news-insights/tracking-tuna-catch-customer



Use Cases

- Phase 1 - the first mile:

- Local fishers receive blockchain keys linked to verified phone and catch data. Simple SMS messages record daily catches and create blockchain assets. When fish are sold, ownership transfers to the supplier, creating a permanent digital history.

Use Cases

• Phase 2 - along the chain:

• Data flow into ERP and supply-chain systems to verify factory inputs and outputs. Smart contracts manage conversion from raw fish to finished products. Blockchain prevents duplicate sustainability claims and allows certifications to be accounted for without exposing every participant's complete dataset.

Use Cases

- Phase 3 - consumer environments:

- Information collected at origin and throughout the supply chain becomes available to downstream buyers. Brands and retailers can replace dense printed communication with mobile-accessible information about producers, suppliers, and processing procedures.

Use Cases

• Zebra Technologies and McCormick & Company:

• McCormick uses Zebra barcode and RFID solutions to trace spices, herbs, and seasonings.

• Ingredients can be tracked from origin to final product, supporting quality and food-safety compliance.

Use Cases

- Fairfood:

- Fairfood is a nonprofit that uses blockchain to provide supply-chain transparency.

- The initiative helps ensure fair compensation for producers and lets consumers trace food to its source.

Use Cases

- Fairfood and coffee producers:

- In collaboration with Koffielust and coffee supplier MAAS, Pure Africa developed an exclusive coffee label called De Idealist.

Use Cases

- Fairfood and coffee producers:

- Pure Africa uses the Trace platform to gain deeper insight into the production chains behind this coffee.

Use Cases

• Fairfood and coffee producers:

• Trabocca is a coffee-bean importer that works with growers, cooperatives, and exporters to supply roasters with traceable green specialty coffees.

Use Cases

- In partnership with Fairfood, Trabocca traced every transaction behind its Ethiopian coffee.

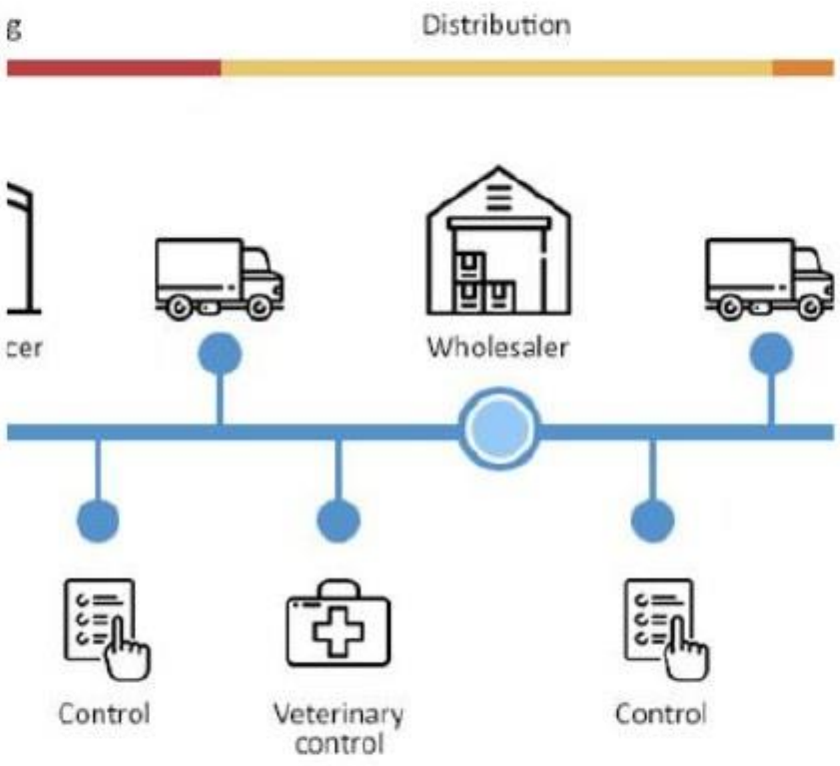
- The data indicated that participating farmers earned 63% more than farmers elsewhere.

Use Cases

• TE-FOOD is a blockchain-based farm-to-table food traceability solution.

• It has been deployed in several countries to track livestock and fresh products.

• In Vietnam, TE-FOOD traces the pork supply chain from breeding to retail, supporting transparency and safety.



Use Cases

• Nestlé and OpenSC:

• Nestlé partnered with blockchain platform OpenSC to trace product supply chains.

• The initiative provides transparency and helps guarantee authenticity and quality.

Use Cases

• Nestlé and OpenSC:

• A traceability tool follows Congo's KAHAWA coffee used in Nespresso capsules.

• https://nestle-nespresso.com/news/new-levels-transparency-coffee-supply-chain
• https://nespresso.opensc.org/product/46155847884615538540/journey

Use Cases

- Nestlé and OpenSC:
- Traceability for Congo's KAHAWA coffee used in Nespresso capsules.

Use Cases

- Nestlé and OpenSC:
- Traceability for Congo's KAHAWA coffee used in Nespresso capsules.

Use Cases

• Carrefour and Hyperledger:

• Carrefour uses Hyperledger Fabric to trace products including meat, milk, and fruit.

• Customers scan a QR code on the package to view the product's journey from farm to shelf.

Use Cases

- Carrefour and BRF:
- BRF and Carrefour partnered with IBM Brazil on the Food Tracking project.
- The goal is blockchain-based product traceability.

Use Cases

- Italian olive oil:

- The Italian government implemented blockchain traceability for olive oil.

- Consumers can trace oil to a specific farm and production batch, helping guarantee authenticity and quality.

Use Cases

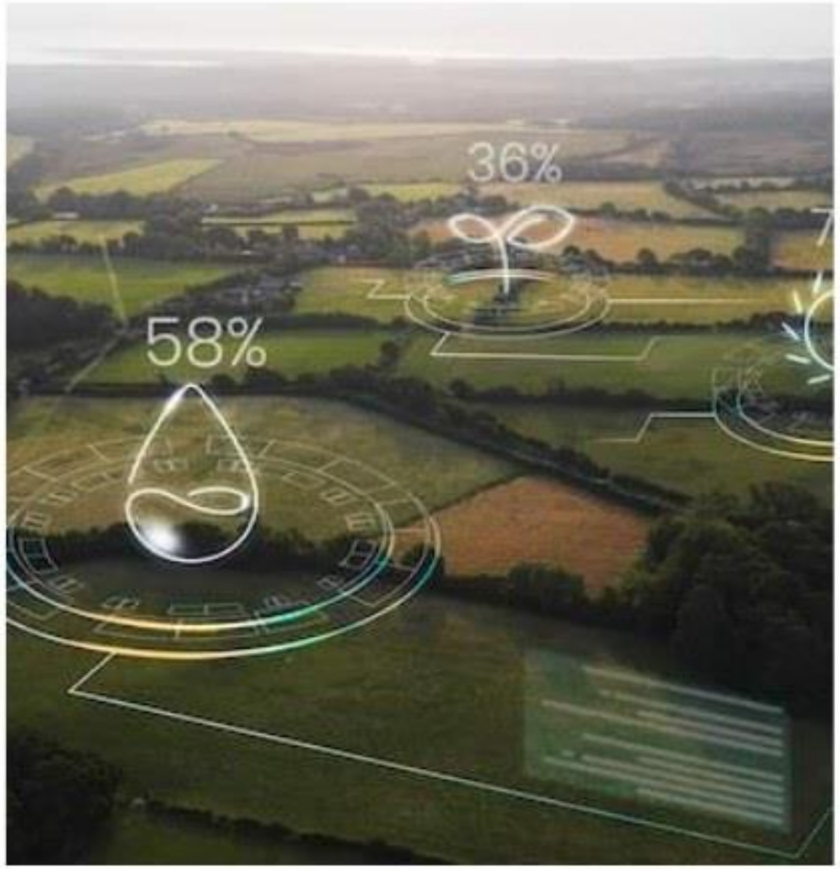
- IBM Food Trust and Coricelli:

- A three-year supply-chain contract covers 2 million kg of Italian olive oil from three organizations in Apulia.
- Coricelli is present in 110 countries; exports represent 54% of sales.
- Four million kg of oil are traced by blockchain.
- The product carries the 100% Italian label 'Signed by Italian Farmers.'

Agriculture 4.0: Big Data

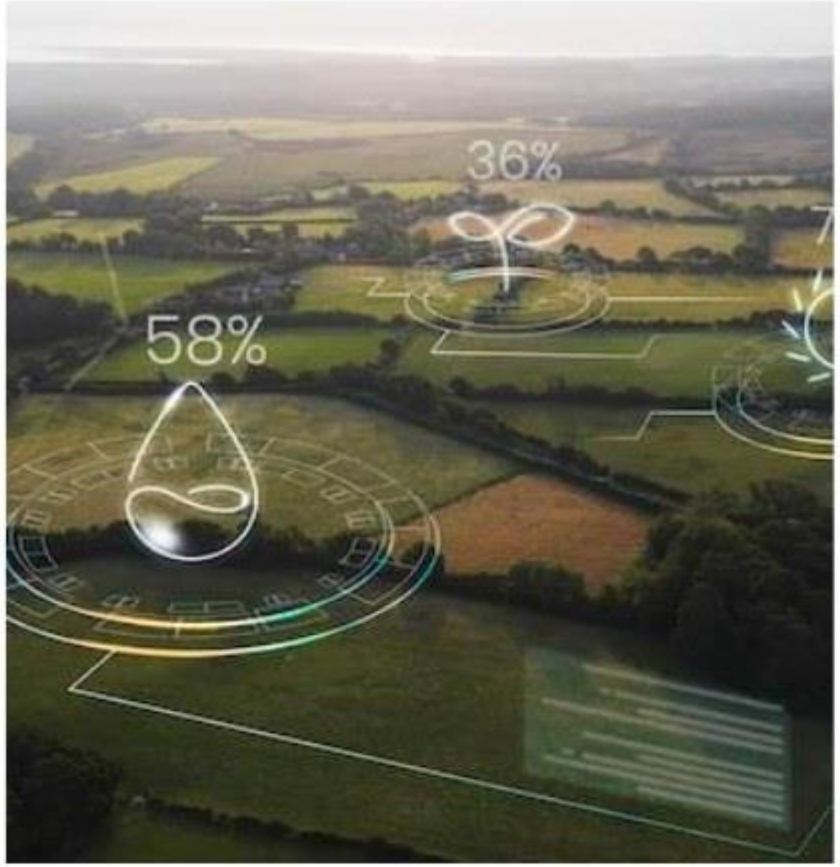
Agriculture 4.0: Big Data

- In Agriculture 4.0, big data refers to the vast and diverse datasets generated by sensors, drones, satellites, weather stations, and agricultural machinery.



Agriculture 4.0: Big Data

• Big-data analytics processes this information to uncover patterns, trends, and correlations, helping farmers make informed decisions.



Agriculture 4.0: Big Data

The five Vs of Big Data:

- Volume: the massive amount of data generated;
- Velocity: how quickly data are collected and processed;
- Variety: multiple data types, including images, text, and numbers;
- Veracity: data quality and reliability;
- Value: the insights and decisions derived from data.

Agriculture 4.0: Cloud Computing

Cloud computing provides on-demand computing resources over the internet. It removes the need to manage physical infrastructure directly and lets users pay only for what they consume.

- Infrastructure as a Service: computing and storage;
- Platform as a Service: development and deployment environments;
- Software as a Service: complete applications delivered as services.

Agriculture 4.0: Cloud Computing

• Cloud computing provides the infrastructure and services needed to store, process, and analyze the large volumes of data generated by agricultural technologies.

Agriculture 4.0: Cloud Computing

- Data storage and management:

- Cloud platforms offer scalable and cost-effective storage for data from sensors, drones, satellites, and agricultural equipment.

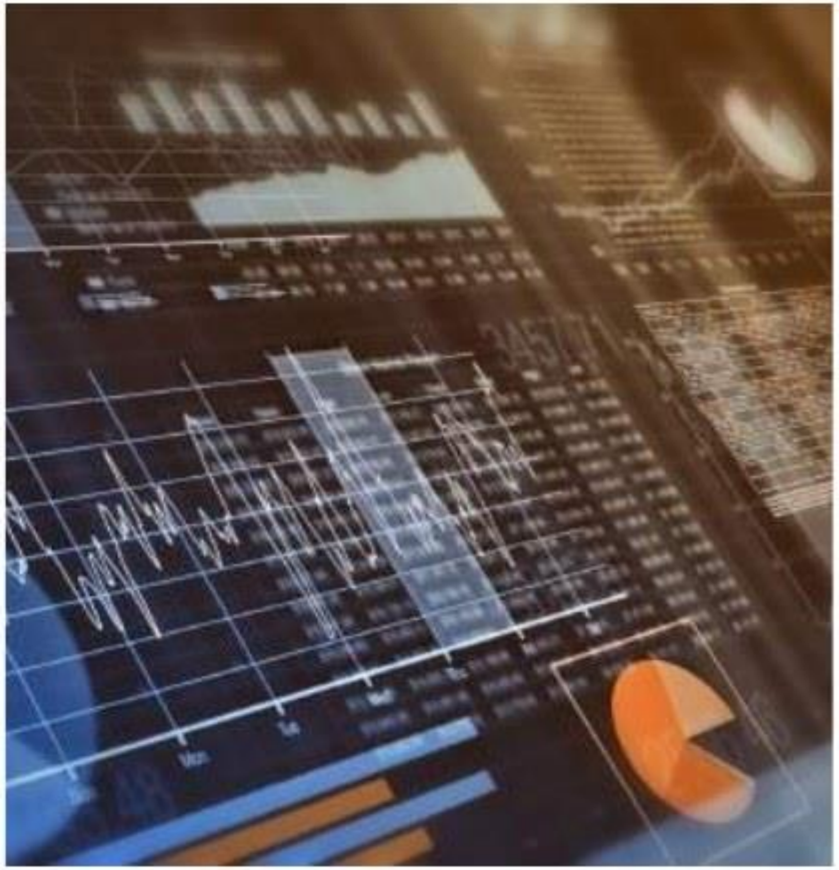
- Datasets may describe soil conditions, crop health, weather patterns, and livestock health.

Agriculture 4.0: Cloud Computing

• Data processing and analytics:

• Cloud computing supplies the processing power needed to extract decision-support insights.

• Machine-learning algorithms can identify patterns, forecast yields, detect crop diseases, and optimize resource allocation.



Agriculture 4.0: Cloud Computing

• Accessibility and collaboration:

• Cloud platforms make agricultural data available to farmers, agronomists, researchers, and other stakeholders wherever internet access exists.

• This supports collaboration, knowledge sharing, faster decisions, and problem solving.

Agriculture 4.0: Cloud Computing

- Scalability and flexibility:

- Agricultural organizations can scale IT resources up or down and pay only for what they use.

- This is especially useful for seasonal operations and unexpected events such as weather changes or pest outbreaks.

Agriculture 4.0: Cloud Computing

• Cost efficiency:

• By reducing the need for expensive local infrastructure and dedicated IT staff, cloud computing can significantly lower agricultural data-storage and processing costs.

Agriculture 4.0: Cloud Computing

• Cloud-computing applications in Agriculture 4.0:

• Farm-management software provides tools for crop planning, field mapping, inventory management, and financial tracking.

Agriculture 4.0: Cloud Computing

- Trimble Ag Software:

- Crop planning;
- Operations monitoring;
- Input tracking;
- Financial reporting.

Agriculture 4.0: Cloud Computing

- Agroptima:

- Agricultural activity records;
- Inventory control;
- Machinery management;
- Reporting and analytics.

Agriculture 4.0: Cloud Computing

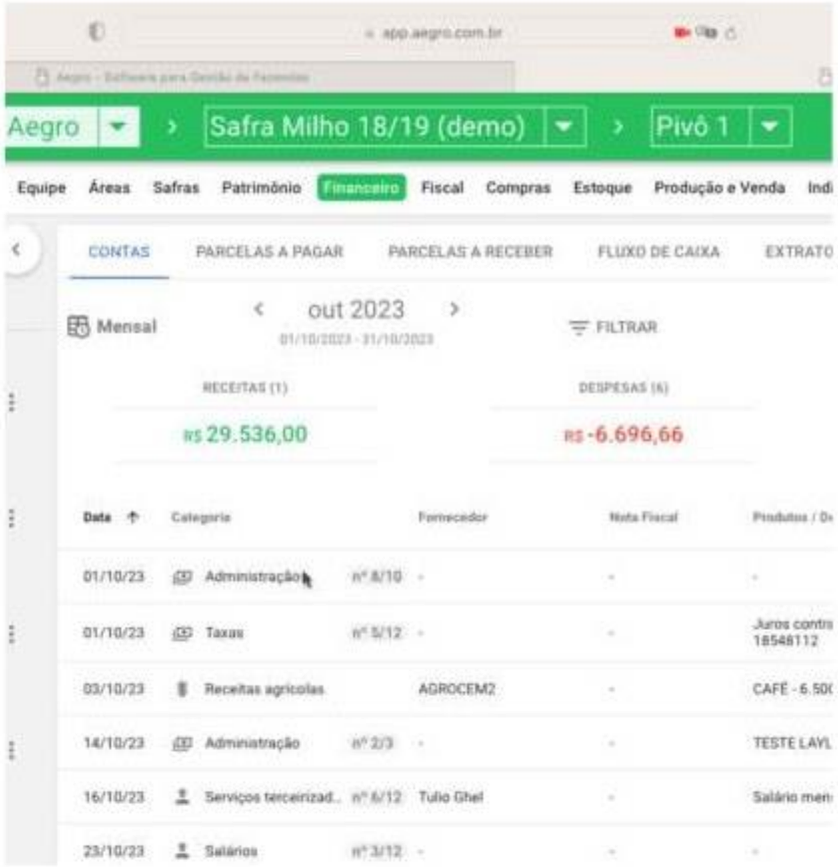
- Granular:

- An agricultural management and analytics platform designed to improve efficiency and profitability.

- Features include crop planning, operations tracking, financial reporting, and team management.

Agriculture 4.0: Cloud Computing

- Aegro is a farm-management platform that supports complete crop planning, from planting-cost budgets to business productivity targets.



Agriculture 4.0: Cloud Computing

• Precision-agriculture platforms use cloud computing to process sensor data and other sources, giving farmers real-time information to optimize irrigation, fertilization, and pest control.

Agriculture 4.0: Cloud Computing

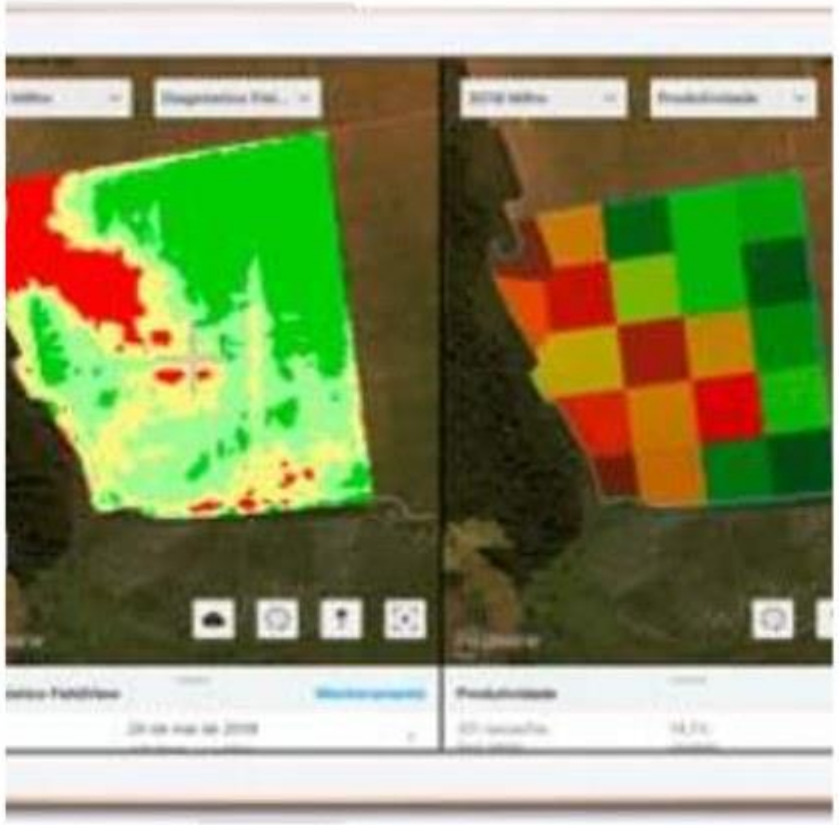
- John Deere Operations Center:

- Machinery monitoring;
- Planting, harvesting, and spraying management;
- Yield mapping.

Agriculture 4.0: Cloud Computing

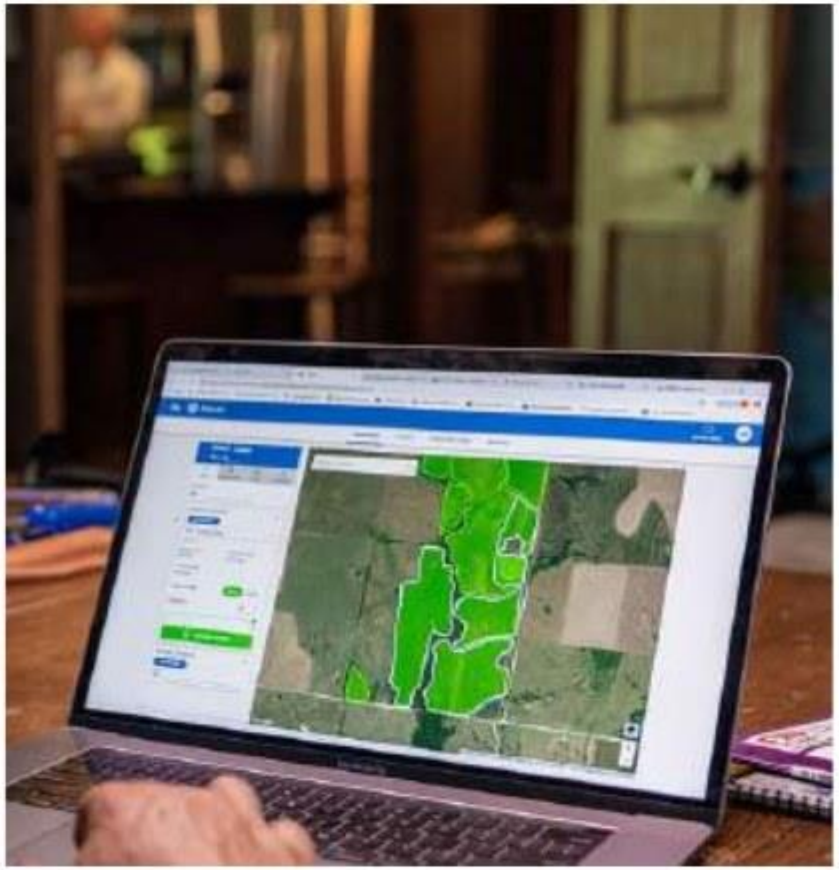
- Climate FieldView:

- A Climate Corporation solution that integrates weather, soil, and crop data to provide real-time insights.



Agriculture 4.0: Cloud Computing

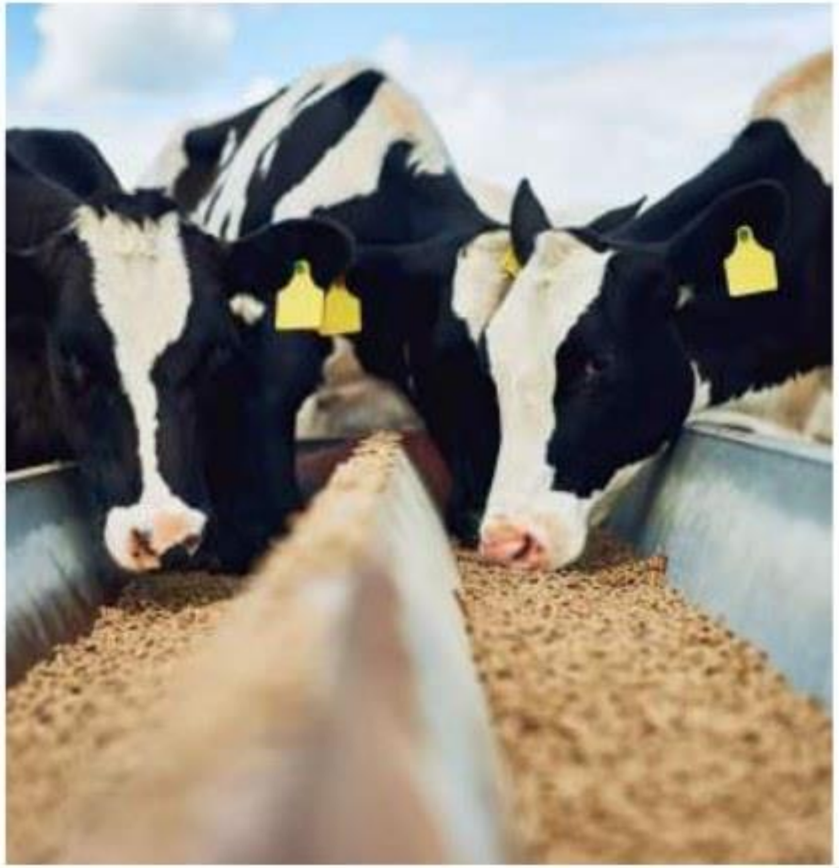
- Topcon Agriculture:
- Precision solutions designed to optimize agricultural operations.



Agriculture 4.0: Cloud Computing

• Livestock-management systems:

• Cloud-based systems monitor animal health, track feeding patterns, and optimize breeding programs.



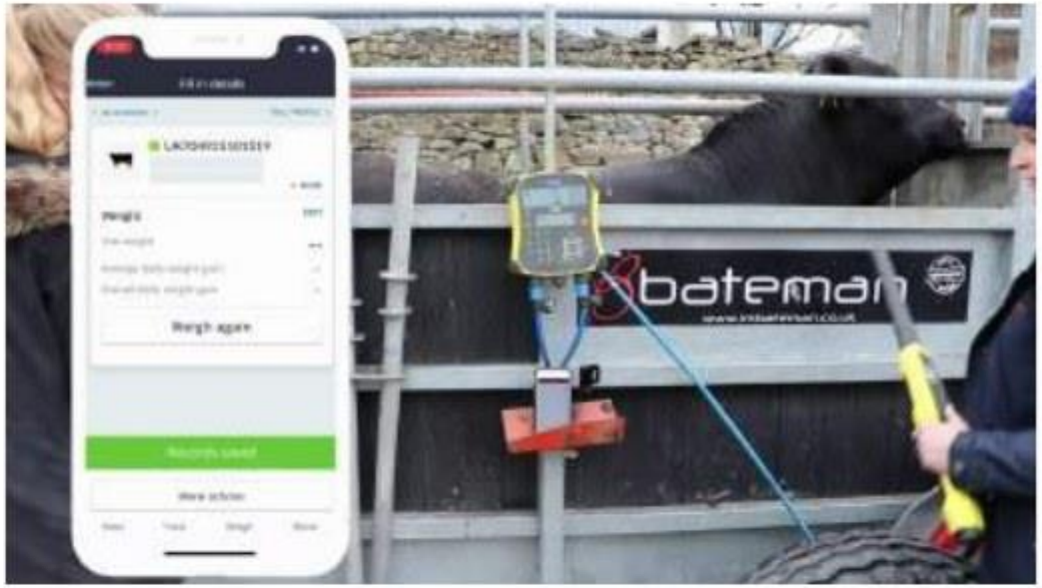
Agriculture 4.0: Cloud Computing

- CattleMax:
- Livestock-management software that helps organize and analyze herd data.

Agriculture 4.0: Cloud Computing

- AgriWebb:

- Livestock tracking;
- Pasture planning;
- Treatment records;
- Performance reporting.



Agriculture 4.0: Cloud Computing

• MooMonitor+:

• A sensor-based system that tracks cattle health and behavior.

• Functions include activity monitoring, heat detection, rumination tracking, and health alerts.

Agriculture 4.0: Cloud Computing

- CowMed:
- A sensor-based monitoring system for dairy-herd management.

Agriculture 4.0: Edge Computing

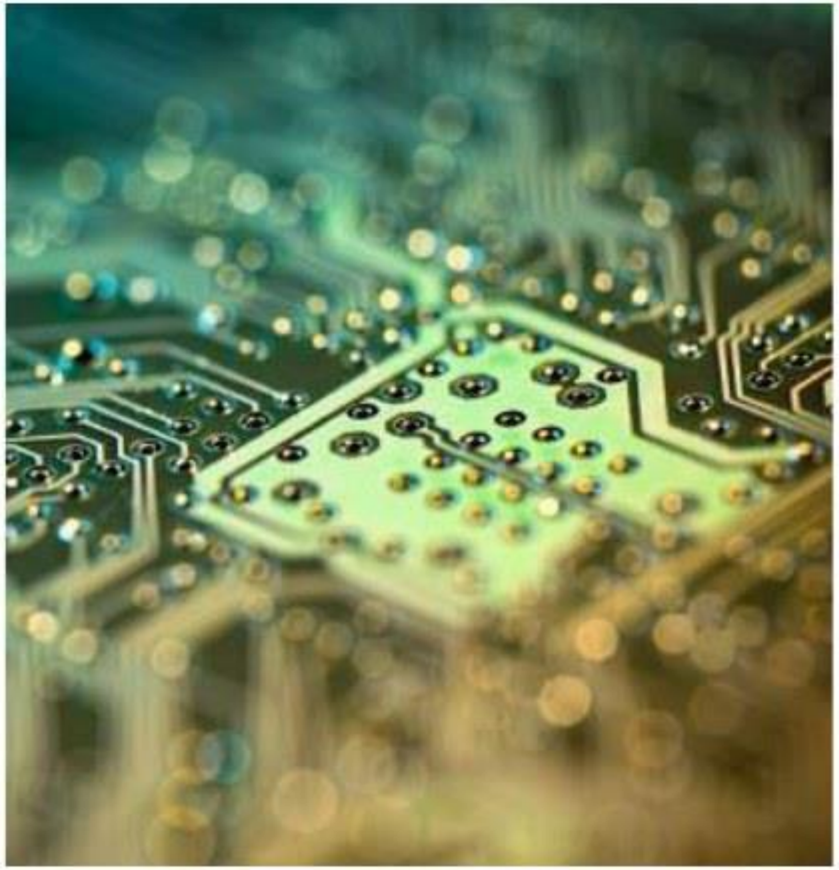
• Edge computing complements the cloud by moving processing closer to where data are generated - at the network edge, within the farm or field.

Agriculture 4.0: Edge Computing

• Edge computing supports Agriculture 4.0 through real-time decisions.

• Processing data at the source reduces latency and enables immediate action.

• Examples include adjusting irrigation from live soil-moisture data or detecting pest infestations early.



Agriculture 4.0: Edge Computing

- Bandwidth optimization:
- Local processing reduces the amount of data transmitted to the cloud, saving bandwidth and cost.
- Only the most relevant processed information needs to be sent for further analysis and storage.

Agriculture 4.0: Edge Computing

- Offline functionality:

- In remote areas with limited or unreliable internet, edge devices can collect data and perform basic analytics autonomously.

- Critical operations can therefore continue without constant connectivity.

Agriculture 4.0: Edge Computing

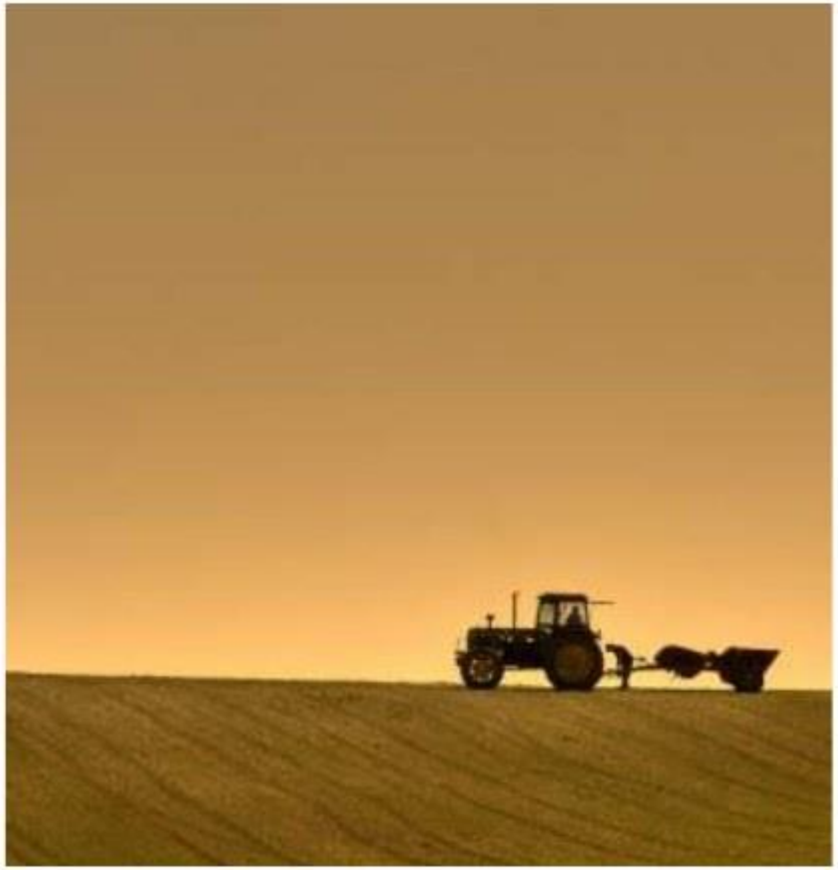
- Improved security:

- Edge computing can keep sensitive agricultural data inside the farm network, reducing exposure during transmission to the cloud.

Agriculture 4.0: Edge Computing

- Scalability and adaptability:

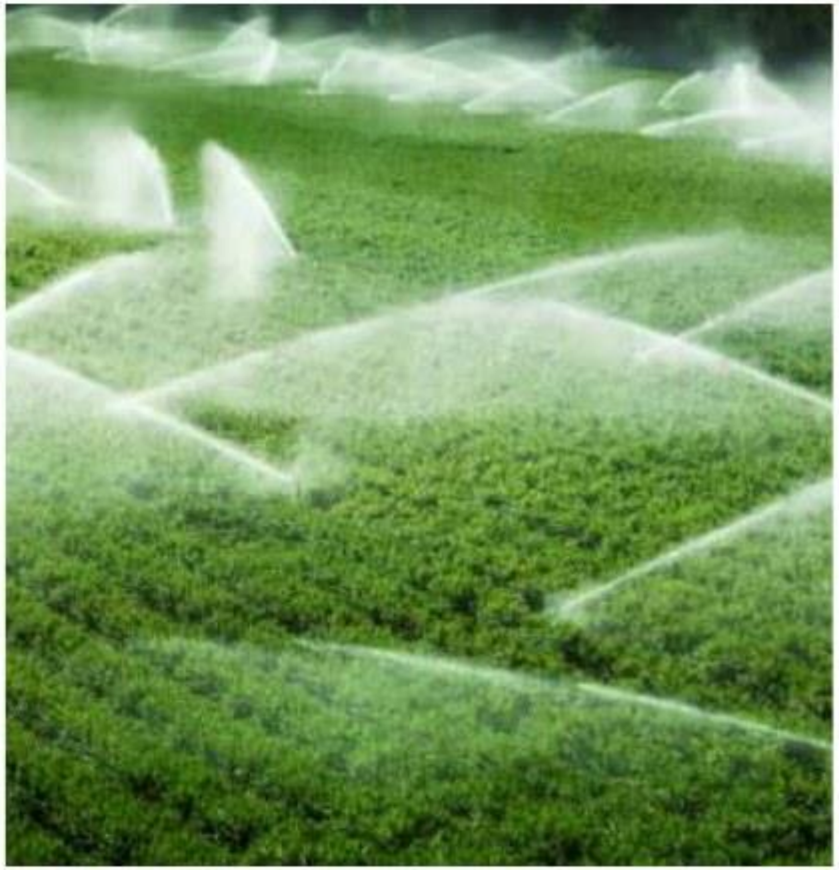
- Edge devices can be deployed and scaled according to the needs of a specific farm or field, providing a flexible solution for many agricultural applications.



Agriculture 4.0: Edge Computing

- Edge-computing applications in Agriculture 4.0 - smart irrigation:

- Edge devices collect soil-moisture and weather-station data, process them locally, and adjust irrigation schedules automatically in real time.



Agriculture 4.0: Edge Computing

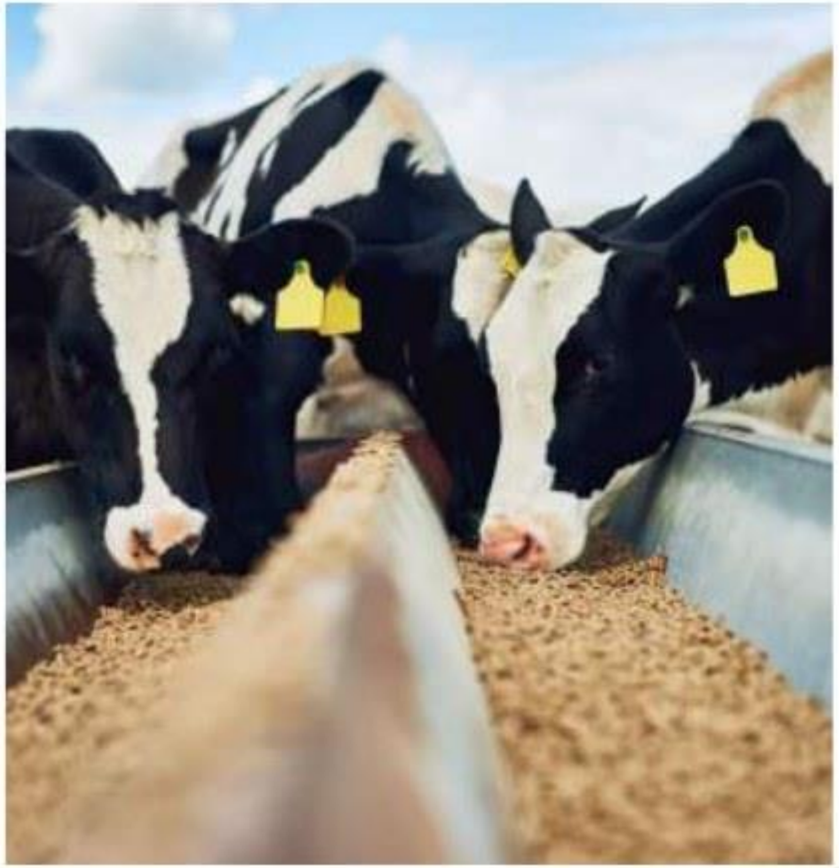
- Autonomous agricultural robots:

- Robots with edge-computing capabilities perform weeding, harvesting, and spraying with minimal human intervention, increasing efficiency and precision.

Agriculture 4.0: Edge Computing

- Livestock monitoring:

- Edge devices monitor animal health and behavior, provide early warnings of potential problems, and support timely intervention.



Agriculture 4.0: Edge Computing

• Crop-disease detection:

• Edge-enabled cameras and sensors analyze plant images for early signs of disease or stress, allowing immediate treatment and helping prevent crop loss.

Big Data Platforms

- Several commercial big-data platforms can support Agriculture 4.0 applications:

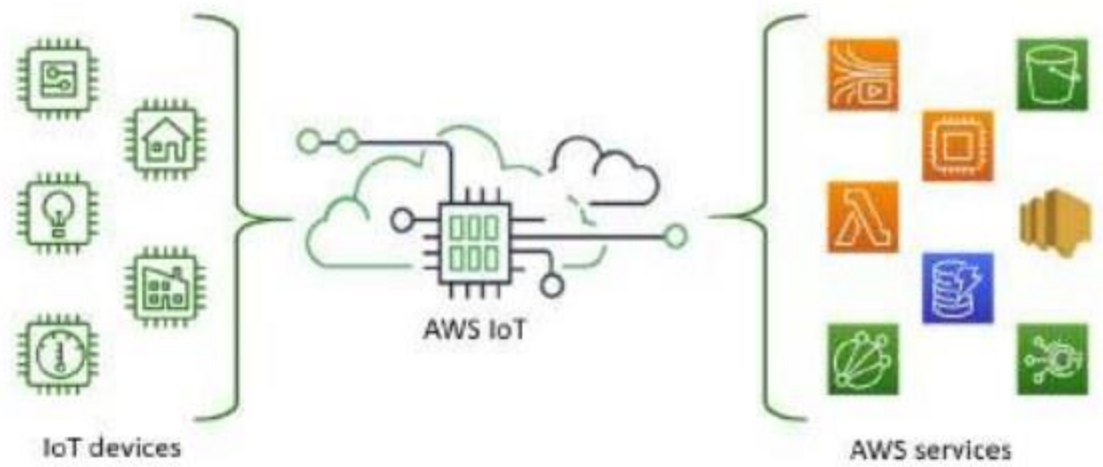
- Amazon Web Services (AWS);
- Microsoft Azure.

- Services may be usage-based, with free tiers available for limited periods or workloads.

Big Data Platforms

• Data collection - Amazon Web Services:

• AWS IoT Core enables connected devices to interact securely and easily with cloud applications.

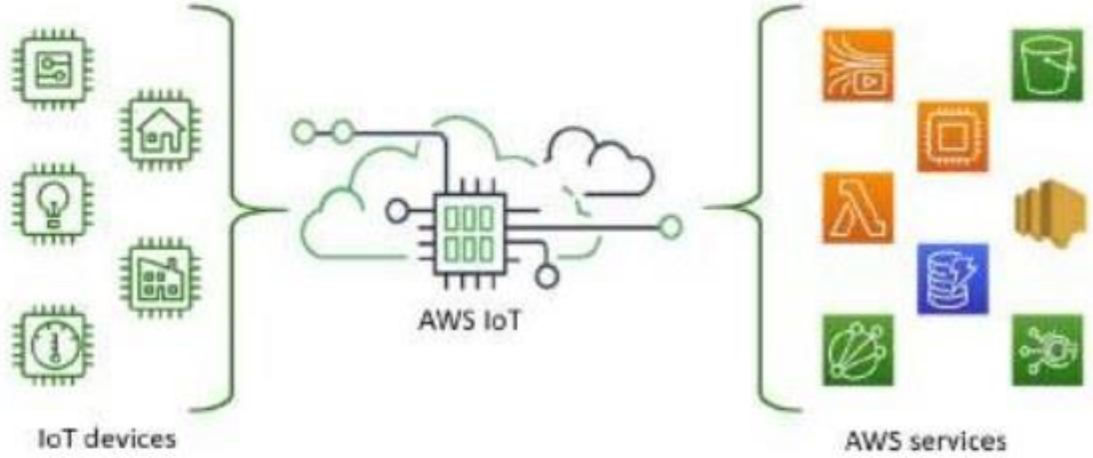


Big Data Platforms

- Data collection:

- AWS IoT Core supports the following protocols:

- MQTT;
- HTTPS;
- LoRaWAN.



Big Data Platforms

• Data processing - Amazon Web Services:

• AWS Lambda executes code in response to events and automatically manages computing resources.

Big Data Platforms

- Data storage - Amazon Web Services:
- Amazon S3 provides scalable storage for large datasets such as drone imagery and sensor readings.
- Amazon RDS is a managed relational-database service for structured data.

Big Data Platforms

- Data analytics - Amazon Web Services:

- Amazon SageMaker is a fully managed service that lets developers and data scientists build, train, and deploy machine-learning models.

Big Data Platforms

- Data analytics:

- Support for major machine-learning frameworks, toolkits, and programming languages.

Big Data Platforms

- Data analytics - Amazon Web Services:

- Amazon Rekognition provides machine-learning-based image and video analysis.

Big Data Platforms: Use Cases

THE CLIMATE CORPORATION

- Amazon Web Services - The Climate Corporation:

- The Climate Corporation uses AWS to collect, process, and analyze large agricultural datasets, including weather and soil information.

Big Data Platforms: Use Cases

- Amazon Web Services:

- AWS IoT Core connects field sensors for real-time monitoring.

- Data are stored in Amazon S3 and analyzed with Amazon Redshift to produce forecasts and recommendations for farmers.

Big Data Platforms: Use Cases

- Amazon Web Services - John Deere:
- John Deere uses AWS to deliver precision-agriculture solutions.

Big Data Platforms: Use Cases

• John Deere equipment uses IoT sensors to collect machine-performance and soil-condition data.

• AWS IoT transmits the data, AWS Lambda and Amazon DynamoDB process them, and Amazon SageMaker builds machine-learning models that optimize operations and improve crop yields.

Big Data Platforms: Use Cases

- Bayer Crop Science:
- Bayer uses AWS to support agricultural research and development.
- Large genomic datasets are stored in Amazon S3.

Big Data Platforms: Use Cases

- Bayer Crop Science:

- Amazon SageMaker builds machine-learning models that identify genetic traits associated with higher yields and disease resistance.

- Amazon Redshift supports rapid data analysis and the development of new crop varieties.

Big Data Platforms: Use Cases

- Farmers Edge:

- Farmers Edge is a digital-agriculture company that uses AWS to provide real-time monitoring and analytics.

- AWS IoT Core connects field sensors and transmits collected data to the cloud.

Big Data Platforms: Use Cases

- Farmers Edge:
- Data are stored in Amazon S3 and processed on Amazon EC2.
- SageMaker predictive models help farmers make irrigation, fertilization, and crop-protection decisions.

Big Data Platforms: Use Cases

- Indigo Agriculture:

- Indigo uses AWS to collect real-time IoT sensor data and optimize plant health and crop yields.

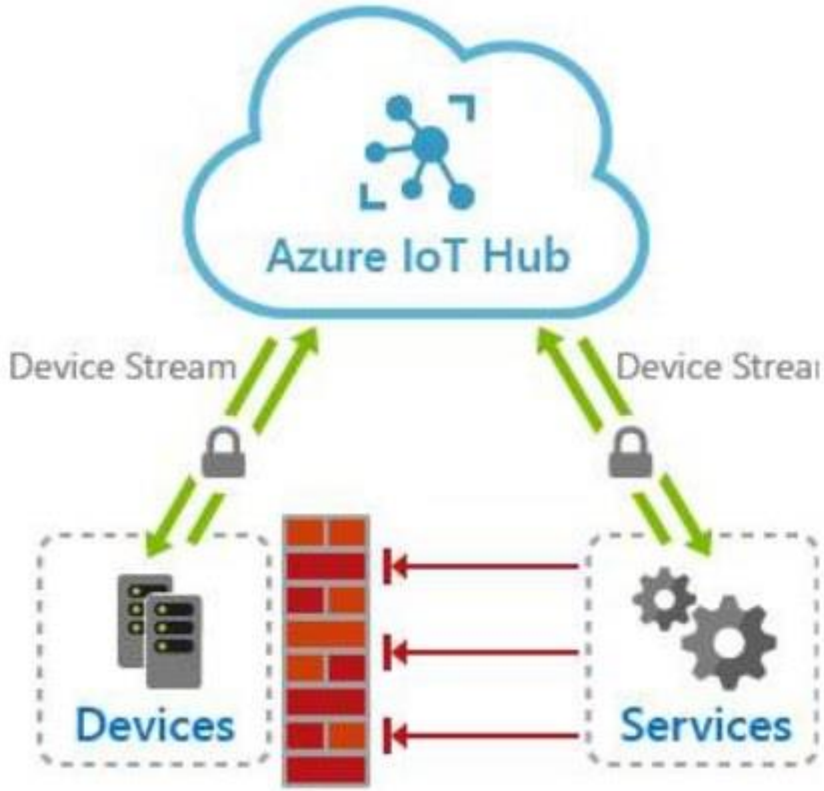
Big Data Platforms: Use Cases

- Indigo Agriculture:
- Amazon Kinesis processes real-time data streams, while Amazon S3 stores the data.
- SageMaker models provide personalized crop-management recommendations.

Big Data Platforms

• Data collection - Azure IoT Hub:

• A managed service for connecting, monitoring, and managing billions of IoT devices.



Big Data Platforms

- Data storage:

- Azure Data Lake Storage;
- Azure SQL Database;
- Azure Cosmos DB.

Big Data Platforms

- Data analytics - Azure Machine Learning:
- A platform for building, training, and deploying machine-learning models.

Big Data Platforms

- Azure Data Manager for Agriculture:

- An agriculture-specific platform that integrates data from sensors, drones, satellite imagery, and third-party sources.

Big Data Platforms

- Agricultural use:

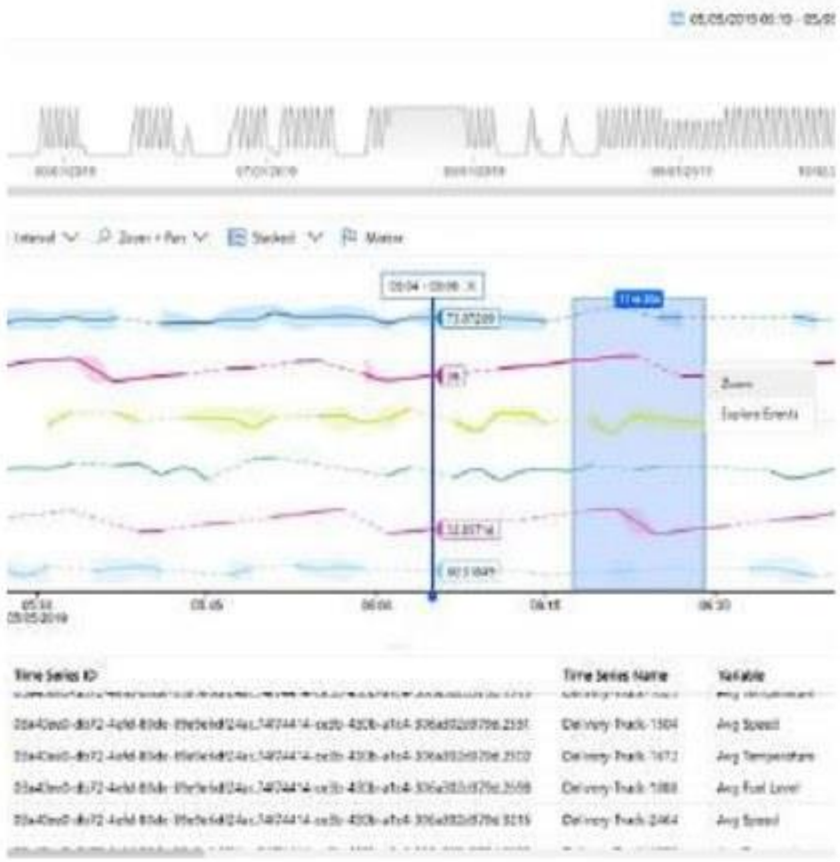
- Collecting and analyzing data from multiple sources to provide detailed insights into farm conditions and support decision-making.

Big Data Platforms

- Azure Time Series Insights:

- A service for time-series analysis and visualization.

- It can monitor farm operations and analyze sensor data over time to identify patterns and anomalies.



Big Data Platforms

- Azure Cognitive Services:
- APIs that add AI capabilities such as image and video analysis.
- Applications include detecting pests, diseases, and water stress through drone and satellite imagery.

Big Data Platforms

- Azure Maps:

- A mapping and geolocation service.

- Applications include agricultural-machinery route planning, field mapping, and product-transport monitoring.

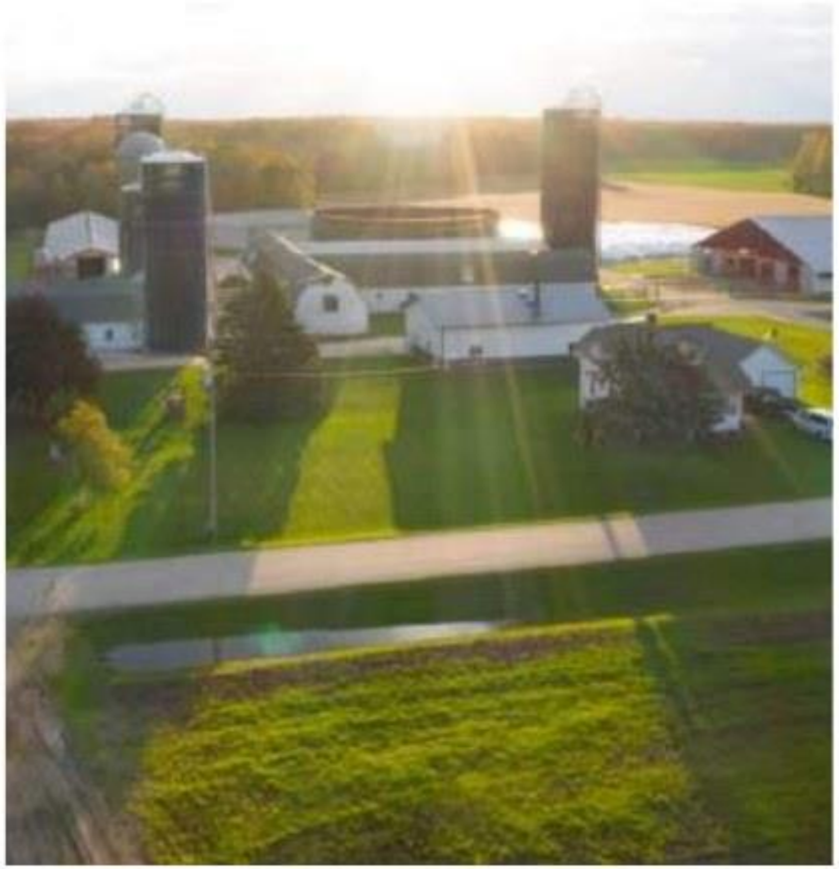
Big Data Platforms

- Azure Synapse Analytics:
- A data-analytics platform that integrates big data and data warehousing.
- It combines multiple agricultural data sources to produce actionable insights.

Big Data Platforms: Use Cases

• Land O'Lakes:

• Uses technology to increase agricultural productivity, optimize resources, and promote sustainability.



Big Data Platforms: Use Cases

- Land O'Lakes:

- Azure Machine Learning forecasts crop yields from historical data and current conditions, enabling more accurate harvest planning and resource management.

Big Data Platforms: Use Cases

- Saturas:

- Improves irrigation efficiency and crop productivity by monitoring plant water needs in real time.

Big Data Platforms: Use Cases

- Saturas:

- In apple orchards, stem-water-potential sensors continuously monitor tree water status.

- Azure IoT Hub and Synapse Analytics provide precise irrigation recommendations, delivering water savings of up to 20% and significant yield improvements.

Recommended Videos

- Land O'Lakes Uses Azure Data Manager for Agriculture to Support Efficient Farming
- https://www.youtube.com/watch?v=kLKB6Zf4jRE

- Building a Unified Agriculture Technology Ecosystem - Amazon Web Services
- https://www.youtube.com/watch?v=NX9pPrD105Y

- All in the Field: AWS Agriculture Live
- https://www.youtube.com/watch?v=S7VLSJgdKCE

References

• Sadiku, M. N. O., Chukwu, U. C. & Sadiku, J. O. (2023). Cloud Computing in Agriculture. World Journal of Agriculture and Urbanization, 2(11), 26-31. https://doi.org/10.51699/wjau.v2i11.11

• Ampatzidis, Y., Partel, V. & Costa, L. Agroview: Cloud-based UAV data analysis for precision agriculture using AI. Computers and Electronics in Agriculture 174, 105457 (2020).

• Punjani, K. K. et al. Cloud computing in agriculture: a bibliometric and network visualization analysis. Quality & Quantity 57, 3849-3883 (2023). https://doi.org/10.1007/s11135-022-01535-1

References

- Amazon Web Services
- https://aws.amazon.com/

- Microsoft Azure
- https://azure.microsoft.com/
- https://azure.microsoft.com/en-us/products/data-manager-for-agriculture